DEXTROSE 50%

Highly irritant — large vein

Many prefer 10–20%

CHEMICAL NAME D-glucose monohydrate ($C_6H_{12}O_6 \cdot H_2O$) 50 g/100 mL hypertonic solution
CLASS Concentrated carbohydrate (hypertonic glucose) solution
TRADE NAMES Glucose 50% · D50W · Dextrose 50% injection

THERAPEUTIC USES / INDICATIONS

- Severe hypoglycaemia with IV access (adults).
- With insulin for hyperkalaemia.
- Hypoglycaemia in insulin overdose; (diluted) parenteral nutrition component.

ADMINISTRATION

- **IV slowly** into a large vein through a large-bore cannula, flushed before and after with saline.
- Hypertonic (≈2,500 mOsm/L) → phlebitis, extravasation necrosis.
- Many guidelines prefer 10% (150–200 mL) or 20% (75–100 mL) glucose over 15 min (safer).
- Children: use 10% glucose only (2 mL/kg).

DOSAGES

- **Adult hypoglycaemia:** 25–50 mL of 50% (12.5–25 g glucose) slowly IV; repeat per glucose.
- **Equivalent safer options:** 10% glucose 150–200 mL or 20% glucose 75–100 mL over 15 min.
- **Hyperkalaemia:** 50 mL 50% glucose (25 g) + 10 units soluble insulin over 15–30 min.
- **Children:** 10% glucose 2 mL/kg (do not use 50%).

PHARMACOKINETICS

- **Onset:** 1–2 min. **Duration:** short — follow with oral/IV carbohydrate.
- Metabolised to CO_2 and water.

PHARMACODYNAMICS

- Directly raises **blood glucose**; with insulin drives K^+ into cells.
- Provides 4 kcal/g energy.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Pain and phlebitis at injection site.
- Hyperglycaemia (overcorrection).
- Feeling of warmth.

ADVERSE EFFECTS (SERIOUS)

- Extravasation → tissue necrosis (severe).
- Hypokalaemia, hypophosphataemia (refeeding).
- Fluid overload, hyperosmolality.
- Rebound hypoglycaemia; Wernicke's encephalopathy in thiamine-deficient patients (give thiamine first).

CONTRAINDICATIONS & CAUTIONS

- Hyperglycaemia.
- Intracranial or intraspinal haemorrhage; delirium tremens with dehydration (relative).
- Anuria (large volumes).

KEY INTERACTIONS

- **Thiamine:** give before glucose in alcohol-dependent/malnourished patients.

NURSING MANAGEMENT

- Confirm cannula patency in large vein; observe site during injection.
- Recheck glucose after 10–15 min; repeat as needed.
- Once conscious, give oral long-acting carbohydrate.
- Monitor K^+ (with insulin) and glucose hourly afterwards.
- Thiamine first in alcohol misuse.

PATIENT EDUCATION

- After recovery, eat a meal or snack to prevent recurrence.
- Report pain or swelling at the drip site.

KEY POINTS

- Hypertonic glucose; rapid rise in 1–2 min.
- Never in children (use 10%).
- Adult 25–50 mL slowly; 10–20% often preferred.
- Extravasation necrosis — large vein, flush.

SOURCES CHECKED **[629]** Glucose 50% w/v Concentrate for solution for infusion (UK SmPC) **[448]** UKKA hyperkalaemia guideline (2023) **[625]** JBDS-IP hypoglycaemia guideline (2018)

Full references p. 432–457